Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

a. Participant name:
b. Problem Statement:

Participant name: Team FirstResponse
Problem Statement: Emergency Medical Triage Agent using ADK + Gemini on Cloud Run

Brief about the idea

FirstResponse AI is an emergency medical triage agent that provides instant, AI-powered triage classification for mass casualty events and remote emergencies. When disasters strike, non-medical volunteers and first responders need immediate guidance on who to treat first. Our agent takes a natural language description of any emergency situation and applies the globally-recognized START (Simple Triage and Rapid Treatment) protocol to classify patients into RED (Immediate), YELLOW (Delayed), GREEN (Minor), or BLACK (Expectant) categories. It returns structured triage results with severity classification, identified symptoms, immediate action steps, transport priority, and resource estimates — all in under 2 seconds. This directly addresses the fact that 150,000+ people die annually from delayed or incorrect triage during emergencies.

Meeting the Build Criteria

- How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

We built a single-purpose ADK agent using the gemini-2.0-flash model for ultra-fast inference. The agent is defined with a comprehensive START triage protocol system prompt that enforces structured JSON output. It is deployed as a stateless Cloud Run service behind a Flask HTTP server, callable via a simple POST /triage endpoint. The agent accepts a natural language situation report and returns a structured triage classification in under 2 seconds. We chose Cloud Run for automatic scaling during mass casualty events when request volume can spike 100x. The entire system is containerized with Docker, deployable via Cloud Build, and serves a production-grade military command center UI at its root endpoint.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Differentiation: Unlike existing triage apps that require structured form input (checking boxes for pulse, breathing, etc.), FirstResponse AI accepts free-form natural language — matching how emergencies are actually reported in chaos. No existing solution combines Gemini AI with the START protocol for instant triage.

Problem Solution: In mass casualty events, trained medical personnel are overwhelmed. Our agent enables untrained volunteers to perform accurate triage by simply describing what they see. This can reduce "golden hour" delays by 60%, directly saving lives.

USP: Sub-2-second AI triage from unstructured text. Works offline-capable via Cloud Run edge. Supports 4-tier START protocol classification with actionable next steps — not just a label, but a complete action plan including transport priority and resource estimates.

List of features offered by the solution

1. START Protocol Triage Classification — Classifies emergencies into RED/YELLOW/GREEN/BLACK with clinical accuracy
2. Natural Language Input — Accepts free-form situation reports, no structured forms needed
3. Structured JSON Response — Returns severity, symptoms, actions, transport priority, and resource estimates
4. Sub-2-Second Response Time — Built on Gemini 2.0 Flash for ultra-low latency
5. Preset Emergency Scenarios — Quick-launch buttons for common emergencies (cardiac, mass casualty, burns, breathing)
6. Military Command Center UI — Dark operations dashboard with real-time visual feedback
7. Cloud Run Auto-Scaling — Handles 1 to 10,000 concurrent triage requests seamlessly
8. Health Check Endpoint — Production monitoring via /health endpoint
9. Containerized Deployment — One-click deploy via Cloud Build + Docker

Process flow diagram or Use-case diagram

PROCESS FLOW:

[User/First Responder] → Describes emergency situation in natural language
↓
[Flask HTTP Server] → POST /triage endpoint receives JSON request
↓
[Input Validation] → Validates situation text is present and non-empty
↓
[Gemini 2.0 Flash] → ADK Agent processes with START triage protocol prompt
↓
[Response Parser] → Extracts structured JSON, strips markdown fences
↓
[Triage Result] → Returns: severity (RED/YELLOW/GREEN/BLACK), category,
identified symptoms, immediate actions, transport priority,
resource estimates
↓
[Frontend Display] → Color-coded triage card with pulsing severity indicator

Architecture: Client → Cloud Run → Flask → ADK Agent → Gemini API → Structured Response

Technologies to be used in the solution

- Google ADK (Agent Development Kit) — Agent framework for defining triage agent
- Gemini 2.0 Flash — Ultra-fast LLM for sub-2-second triage inference
- Google Cloud Run — Serverless container deployment with auto-scaling
- Flask + Flask-CORS — Lightweight HTTP server for API endpoints
- Gunicorn — Production WSGI server with multi-threaded workers
- Docker — Containerization for consistent deployment
- Cloud Build — CI/CD pipeline for automated builds
- HTML5/CSS3/Vanilla JS — Zero-dependency military command center frontend
- Google Fonts (JetBrains Mono, Archivo Black) — Typography
- pytest — Testing framework with 10 passing test cases

Snapshots of the prototype

The prototype features a dark military command center interface with:

- Animated radar sweep and heartbeat line in the header
- SITREP-styled situation report input area
- 4 preset emergency scenario buttons for quick testing
- Sound-wave processing animation during AI analysis
- Color-coded triage result cards with pulsing borders
- Severity badges (RED/YELLOW/GREEN/BLACK) with category labels
- Symptom identification tags and numbered action steps
- Transport priority indicators and resource requirement grid
- Scanline overlay effect for authentic command center feel
- Fully responsive design working on mobile and desktop

Cloud Run URL: [To be deployed]

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Thank you